Ch2. Entropy, Relative Entropy, and mutual information

2.1 Entropy

↳ Measure of the uncertainty of a random variable.

X : a discrete random variable.

$x \in \mathcal{X}$. ($\mathcal{Y}$)

$p(x) = \Pr(X = x)$.

Def Entropy of X : $H(X) := -\sum_{x \in \mathcal{X}} p(x) \log p(x)$.

$* \; p(x) = 0 \Rightarrow \boxed{\lim_{y \to 0+} y \log y = 0}$

↳ $(0 \log 0 = 0)$.

Lemma $H(X) \geq 0$. $\quad p(x) \leq 1$.

(pf) $H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x) = \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{p(x)}$

≥ 0. □

Ex. $X = \begin{cases} 1 & \text{with prob } p \\ 0 & \text{"} \quad 1-p. \end{cases}$

$\Rightarrow H(X) = \boxed{-p \log p - (1-p) \log (1-p) =: H(p)}$.

2.2. Joint Entropy & Conditional Entropy.

Def · Joint entropy

$$H(X,Y) = -\sum_{\substack{x\in\mathcal{X}\\ y\in\mathcal{Y}}} p(x,y)\log p(x,y)$$

$$\boxed{* \ p(x,y) = \Pr(X=x, Y=y)}$$

· $H(Y|X) = \sum_{x\in\mathcal{X}} p(x)\, H(Y|X=x)$.

$$= \sum_{x\in\mathcal{X}} p(x)\left(-\sum_{y\in\mathcal{Y}} p(y|x)\log p(y|x)\right)$$

$*$ $p(y|x)$: conditional probability

$$p(y|x) = \frac{p(x,y)}{p(x)}$$

(Ex) ① $X=Y \sim p(x)$.

$$\Rightarrow H(Y|X) = \sum_{x\in\mathcal{X}} p(x)\left(-\sum_{y\in\mathcal{Y}} p(y|x)\log p(y|x)\right)$$

$$p(y|x) = \begin{cases} 1 & x=y \\ 0. & x\neq y. \end{cases} = \delta_{xy}$$

$x = X.$

$\Rightarrow Y = x$

$\delta_{xy} = 0 \sim 1$

$$= \sum_{x\in\mathcal{X}} p(x) - \sum_{y\in\mathcal{Y}} \delta_{xy}\log\delta_{xy} = 0$$

□

② X, Y are Independent.

$$\boxed{p(x,y) = p(x)p(y)}$$

$$H(Y|X) = -\sum_{x,y} p(x)p(y)\log p(y) = -\sum_{y} p(y)\log p(y) = H(Y)$$

Thus $H(X,Y) = H(X) + H(Y|X)$.

(pf) $H(X) + H(Y|X)$

$$= -\sum_{x} p(x)\log p(x) - \sum_{x,y} p(x,y)\log p(y|x)$$

$$= -\sum_{x,y} p(x,y)\log p(x) - \sum_{x,y} p(x,y)\log p(y|x)$$

$$= -\sum_{x,y} p(x,y)\log p(x,y) = H(X,Y)$$ □

Cor. $H(X,Y|Z) = H(X|Z) + H(Y|X,Z)$.

$$-\sum_{x,y,z} p(x,y,z)\log p(x,y|z) \qquad -\sum_{x,y,z} p(x,y,z)\log p(y|x,z)$$

$$-\sum_{x,z} p(x,z)\log p(x|z)$$

□

2.3 Relative Entropy and mutual Information.

(Def). Relative Entropy (Kullback-Leibler distance)

$\Rightarrow D(p\|q) = \sum_{x} p(x) \log \left(p(x) / q(x) \right)$.

$\ast\ D(p\|q) \neq D(q\|p)$.

$\{ 0 \log 0 = 0 \}$

Remark if $\exists\ \tilde{x} \in \mathcal{X}$ s.t. $p(\tilde{x}) > 0$, $q(\tilde{x}) = 0$

$\Rightarrow \boxed{D(p\|q) = \infty}$

$\left(p(\tilde{x}) \log \left(p(\tilde{x}) / \underset{0^+}{q(\tilde{x})} \right) = \infty \right)$

(Def) Mutual Information

$I(X;Y) = \sum_{x,y} p(x,y) \log \dfrac{p(x,y)}{p(x)p(y)}$

(Ex) if $X, Y \rightarrow$ independent $p(x,y) = p(x)p(y)$.

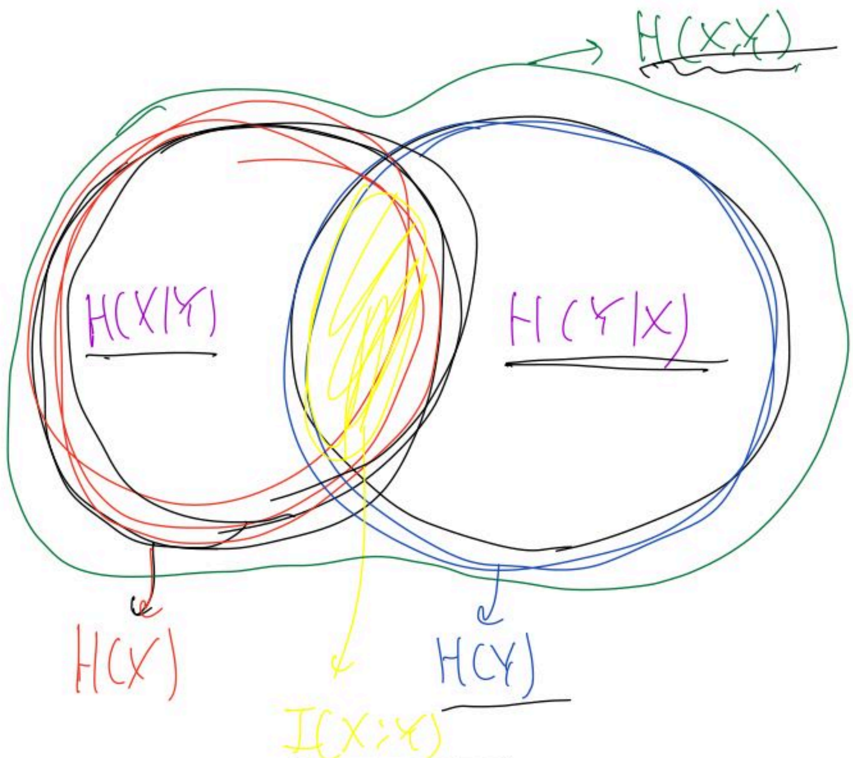
$\Rightarrow \boxed{I(X;Y) = 0}$.

2.4. Relationship between Entropy, ...

$$I(X;Y) = H(X) - H(X|Y).$$

(Pf) —— . Omit.

Thm 2.4.1 Mutual information and Entropy.



e.g. • $H(X,Y) = H(X|Y) + I(X;Y) + H(Y|X).$

• $H(X) = H(X|Y) + I(X;Y)$

2.5. Chain Rules for entropy, relative entropy.

$\boxed{\text{Thm}}$ Chain Rule for <u>Entropy</u>

$$H(X_1, X_2, \cdots, X_n) = \sum_{i=1}^{n} H(X_i | X_{i-1}, \cdots X_1).$$

(pf) . ______ ☒

(Conditional mutual Information)

$\boxed{\text{Def}}$ $$I(X;Y|Z) := \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)}$$

$$= \boxed{H(X|Z) - H(X|Y,Z)}$$

$\boxed{\text{Thm}}$ Chain Rule for <u>Information</u>.

$$I(X_1, X_2, \cdots, X_n ; Y)$$
$$= \sum_{i=1}^{n} I(X_i ; Y | X_{i-1}, \cdots, X_1).$$

$\boxed{\text{Def}}$ Conditional <u>relative entropy</u>.

$$D(p(y|x) \| q(y|x)) = \sum_{x,y} p(x)p(y|x) \log \frac{p(y|x)}{q(y|x)}.$$

$\boxed{\text{Thm}}$ Chain Rule for <u>relative entropy</u>.

$$D(p(x,y) \| q(x,y)) = D(p(x) \| q(x)) + D(p(y|x) \| q(y|x)).$$

2.6. Jensen's Inequality and its consequences.

Def. ① convex ftn : $f(\lambda x_1 + (1-\lambda)x_2) \le \lambda f(x_1) + (1-\lambda) f(x_2)$ $\forall \lambda \in [0,1]$, $x_1, x_2 \in X$

② strictly (" $<$ ") : $\boxed{\text{"}=\text{"} \Leftrightarrow \lambda = 0 \text{ or } 1}$

③ concave ftn : $-f$ is convex ftn $\Rightarrow f$ is concave ftn)

Thm (Jensen's Ineq). f : convex.

$\ast$ $E(f(X)) \ge f(EX)$

$\boxed{\text{expectation value}}$ (pf) omit.

Thm 2.6.3. (Information Inequality).

$D(p\|q) \ge 0$ $\boxed{(\text{equality} : p = q)}$

(pf) $-D(p\|q) = -\sum_{x \in A} p(x) \log p(x)/q(x)$

$\mathcal{X} \supseteq A = \mathrm{supp}(p)$

(if $x \in A \Leftrightarrow p(x) \neq 0$).

$$= \sum_{x \in A} p(x) \log \left(\frac{q(x)}{p(x)}\right)$$

$$\le \log \left(\sum_{x \in A} p(x) \cdot \frac{q(x)}{p(x)} \right)$$

$$= \log\left(\sum_{x\in A} q(x)\right) \le \log\left(\sum_{x\in \mathcal{X}} q(x)\right) = \log 1 = 0.$$

$(A \subseteq \mathcal{X})$

$$\boxed{\therefore\ D(p\|q) \ge 0.}$$

Cor ① $I(X;Y) = D(p(x,y)\|p(x)p(y)) \ge 0.$

② $D(p(y|x)\|q(y|x)) \ge 0.$

③ $I(X;Y|Z) = D(p(x,y|z)\|p(x|z)p(y|z)) \ge 0.$

$\boxed{*\ H(X) \ge 0}$

Thm $H(X) \le \log|\mathcal{X}|$. (the number of elements in the range of X $\mathcal{X}$.)

(pf) $u(x) = 1/|\mathcal{X}|$.

$$D(p\|u) = \sum_x p(x)\log\frac{p(x)}{u(x)}$$

$0 \le$ (Inf. Ineq.) $= \sum_x p(x)(\log p(x) + \log|\mathcal{X}|)$

$$= -H(X) + 1\cdot\log|\mathcal{X}|.$$

$$\Longrightarrow \boxed{H(X) \le \log|\mathcal{X}|.}$$

Thm (Conditioning reduces entropy).

$$\boxed{H(X|Y) \leq H(X)}$$

(pf) $0 \leq I(X;Y) = H(X) - H(X|Y)$

$\Rightarrow H(X) \geq H(X|Y)$. ☐

Thm (Independence bound on entropy).

$$H(X_1, X_2, \cdots, X_n) \leq \sum_{i=1}^{n} H(X_i).$$

(pf) $H(X_1, \cdots, X_n) = \sum_{i=1}^{n} H(X_i | X_{i-1}, \cdots, X_1)$.

Chain rule. $\leq \sum_{i=1}^{n} H(X_i)$ ☐

9:55 Begin.

2.7. Log sum Inequality and its application

Thm (Log sum Inequality)

$a_1, \cdots, a_n, b_1, \cdots, b_n \geq 0$.

$$\Rightarrow \boxed{\sum_{i=1}^{n} a_i \log \frac{a_i}{b_i} \geq \sum_{i=1}^{n} a_i \log\left(\frac{\sum_{i=1}^{n} a_i}{\sum_{i=1}^{n} b_i}\right)}$$

(pf) (Key pt: Jensen's Ineq.)

Thm (Convexity of relative entropy)

$D(p\|q)$ is convex in the pair (p, q).

i.e. $D(\lambda p_1 + (1-\lambda)p_2 \| \lambda q_1 + (1-\lambda) q_2) \le \lambda D(p_1\|q_1) + (1-\lambda) D(p_2\|q_2)$.

(pf) L.H.S. $= \sum_x (\lambda p_1(x) + (1-\lambda) p_2(x)) \log \frac{\lambda p_1(x) + (1-\lambda) p_2(x)}{\lambda q_1(x) + (1-\lambda) q_2(x)}$

$$\le \sum_x \left(\lambda p_1(x) \log \frac{\lambda p_1(x)}{\lambda q_1(x)} + (1-\lambda) p_2(x) \log \frac{(1-\lambda) p_2(x)}{(1-\lambda) q_2(x)} \right)$$

$$= \lambda D(p_1\|q_1) + (1-\lambda) D(p_2\|q_2)$$

QED

Thm $H(p)$ is a concave ftn of p.

(pf). $H(p) = \underbrace{\log|\mathcal{X}|}_{\text{const}} - D(p\|u)$

uniform

(p_1, u), (p_2, u).

QED

Thm (1) $I(X;Y)$ is concave ftn of $p(x)$ for fixed $p(y|x)$

(2) " $p(y|x)$ " $p(x)$

2.8. Data-processing Inequality

Def. X, Y, Z form a Markov chain $X \to Y \to Z$

iff $p(x,y,z) = p(x)\,p(y|x)\,p(z|y)$.

$$\xrightarrow{p(x)} x \xrightarrow{p(y|x)} y \xrightarrow{p(z|y)} z$$

Thm 2.8.1 (Data processing inequality) $X \to Y \to Z$ Markov chain

$$\Rightarrow \boxed{I(X;Y) \geq I(X;Z)}$$

(pf) $I(X;Y,Z) = I(X;Z) + I(X;Y|Z)$

$= I(X;Y) + I(X;Z|Y)$

$\hookrightarrow I(X;Z|Y) = \sum_{x,y,z} p(x,y,z) \log \left(\dfrac{p(x,z|y)}{p(x|y)\,p(z|y)} \right)$

$\left(x, z \text{ are indep} \right) = 1 \quad \cdots\cdots \quad = 0.$

$$\boxed{I(X;Y) = I(X;Z) + I(X;Y|Z)}$$

$\hookrightarrow I(X;Y) \geq I(X;Z)$ ☑

Coro ① $Z = g(Y) \Rightarrow I(X;Y) \geq I(X;g(Y))$

$\geq I(X; g^{-1}(g(Y)) = Y)$

② $I(X;Y) \geq I(X;Y|Z)$.

2.9. Sufficient Statistics.

We consider $\{f_\theta(x)\}$: indexed by θ.

$$\theta \longrightarrow X \longrightarrow T(X)$$

any statistic (mean, variance, ...).

Def. ① T is sufficient statistic relative to $\{f_\theta(x)\}$

if $\boxed{\theta \longrightarrow T(X) \longrightarrow X}$ forms a Markov chain

② T is minimal sufficient statistic relative to $\{f_\theta(x)\}$ if $\theta \rightarrow T(X) \rightarrow U(X) \rightarrow X$

forms a Markov chain $\forall U$: sufficient statistic

e.g. $X_1, \cdots, X_n$, $X_i \in \{0, 1\}$

i.i.d. (independent identically distributed ...)

$\theta = \Pr(X_i = 1)$,

$T(X_1, \cdots, X_n) = \sum_{i=1}^n X_i$

$$\Pr\left((X_1, \cdots, X_n) = (x_1, \cdots, x_n) \,\middle|\, \sum_{i=1}^n X_i = k\right) = \begin{cases} 1/\binom{n}{k} & \sum x_i = k \\ 0 & \text{otherwise.} \end{cases}$$

$$\theta \longrightarrow \sum X_i \longrightarrow (X_1, \cdots, X_n)$$

indep θ.

2.10. Fano's Ineq.

$$X \to Y \to \hat{X}.$$

estimate.

$\boxed{\text{Thm 2.10.1}}$ (Fano's Inequality)

For any estimator $\hat{X}$ s.t. $X \to Y \to \hat{X}$.

with $P_e = \Pr(X \neq \hat{X})$, we have

$$H(P_e) + P_e \log|\mathcal{X}| \geq H(X|\hat{X}) \geq H(X|Y).$$

Rmk $P_e = 0 \iff H(X|Y) = 0$

(Same distribution).

$\boxed{\text{Pf of Fano's inequality}}$

$$E = \begin{cases} 1 & \hat{X} \neq X \to P_e \\ 0 & \hat{X} = X. \to 1-P_e \end{cases}$$

$$H(E, X|\hat{X}) = H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{\to 0}$$

Chain

$$= H(E|\hat{X}) + H(X|E, \hat{X})$$

Conditioning reduces entropy

$$\leq H(P_e) + P_e \log|\mathcal{X}|$$

$\Rightarrow$ $H(X|E,\hat{X}) = \Pr(E=0)H(X|\hat{X}, E=0)$
$+\Pr(E=1)H(X|\hat{X}, E=1)$
$\le (1-P_e)\cdot 0 \quad + P_e \log|\mathcal{X}|$

$\Rightarrow$ $H(X|\hat{X}) \le H(E,X|\hat{X}) \le H(P_e) + P_e\log|\mathcal{X}|$.

$\Rightarrow H(P_e) + P_e\log|\mathcal{X}| \ge H(X|\hat{X}) \ge H(X|Y)$

$\boxed{X \to Y \to \hat{X}}$

∎

$\boxed{\text{Cor}}$ ① $\hat{X} = Y \Rightarrow H(p) + p\log|\mathcal{X}| \ge H(X|Y)$

$\boxed{p = \Pr(X \ne Y)}$.

② $H(P_e) + P_e\log(|\mathcal{X}|-1) \ge H(X|Y)$.

$\boxed{\text{Lem}}$. X, X' : i.i.d with entropy $H(X)$.

$\Pr(X = X') \ge 2^{-H(X)}$. ("=" $\Leftrightarrow$ X : uniform).

(pf). $2^{E\log p} \le E(2^{\log p})$. ($\log x = \log_2 x$).

$\Rightarrow 2^{-H(X)} \le \sum_x p(x) 2^{\log p(x)} = \sum_{x\in\mathcal{X}} p(x)^2$

$= \Pr(X = X')$. ∎

Cor. X, X': Independent $X \sim p(x)$ $\boxed{x, x' \in \mathcal{X}}$

$X' \sim r(x')$.

$$\Rightarrow \boxed{Pr(X = X') \le 2^{-H(p) - D(p\|r)}}$$

$$Pr(X = X') \le 2^{-H(r) - D(r\|p)}.$$

(Pf.)

$$2^{-H(p) - D(p\|r)} = 2^{\sum (p\log p + p\log \frac{r}{p})}$$

$$= 2^{\sum p\log r} \overset{?}{\le} \sum p 2^{\log r}$$

$$= \sum_x p(x) r(x)$$

$$= Pr(X = X')$$

証